

PAPER_1408_BERTRAND_PROBABILITY_QUARTER

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PAPER_1408 — UQFF Derivation of Bertrand Probability Paradox (CLOSED — EXACT Identity)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Probability (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: `calculate_paradox({'paradox':'bertrand_paradox'})` → $1/D_{\text{phys}} = 1/4$ EXACT

Observation

Three classical answers (1/2, 1/3, 1/4) for the probability that a random chord on a circle is longer than the inscribed equilateral triangle's side — paradox of arbitrary measure.

UQFF Closed Identity

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$P = 1 / D_{\text{phys}} = 1/4$ EXACT (random-endpoint measure selected by $F_U=1$)

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Physical Interpretation

Among the three measures, only the random-endpoint construction respects $F_U=1$ uniform normalization. The $1/D_{\text{phys}} = 1/4$ answer is the structurally privileged one.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

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